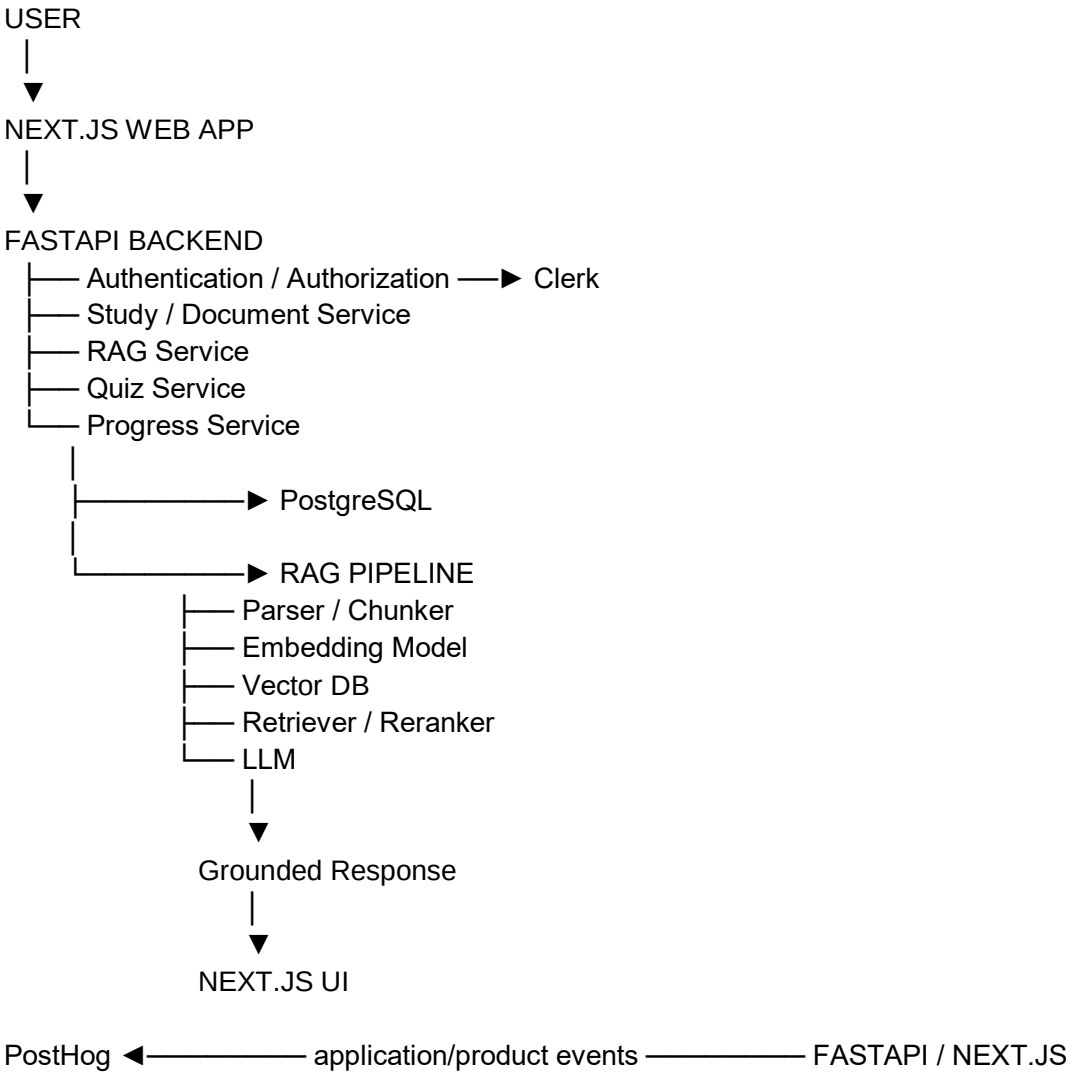
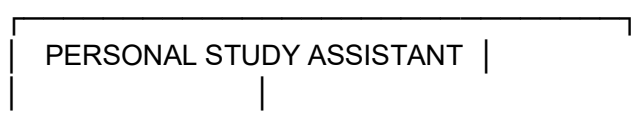
Personal Study Assistant

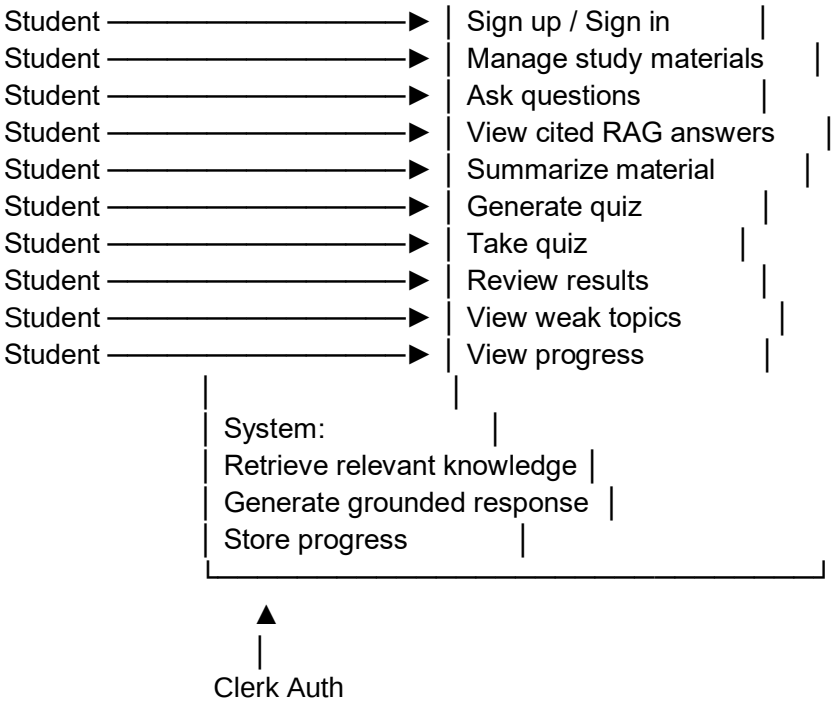
System Design, UML & Application Diagrams

1. System Context / High-Level Architecture

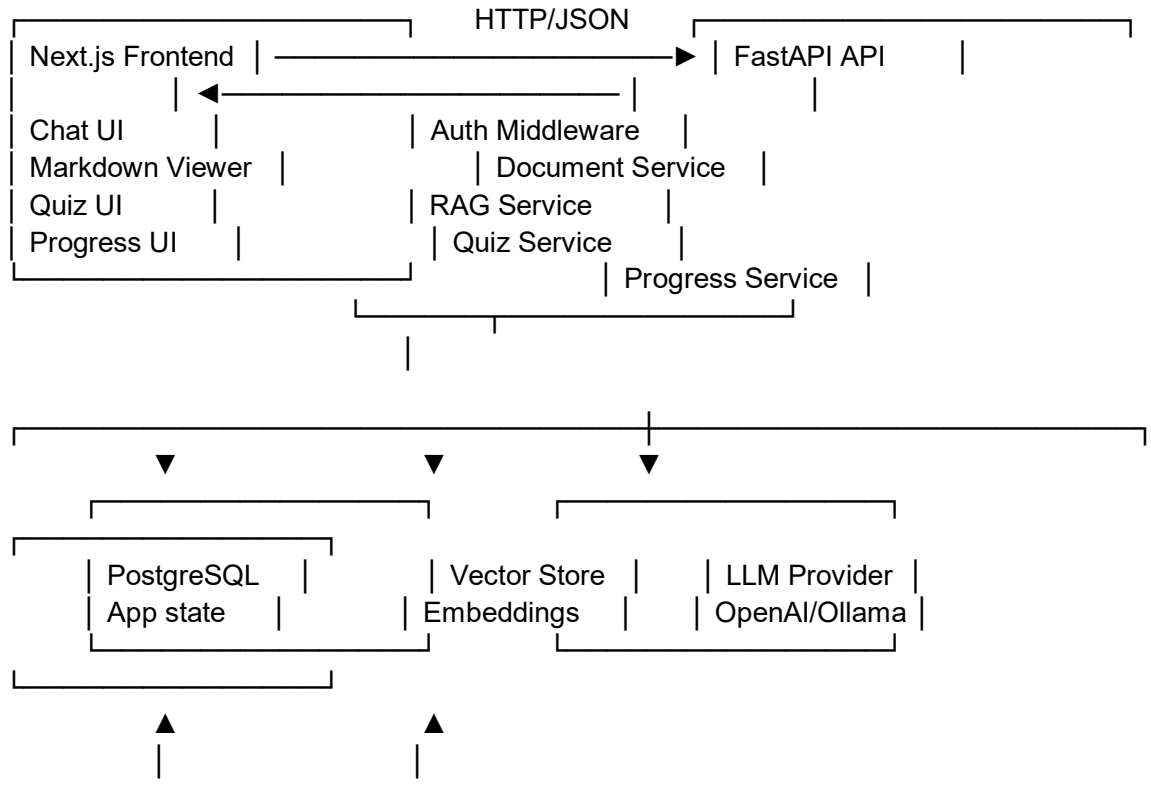


2. Use Case Diagram



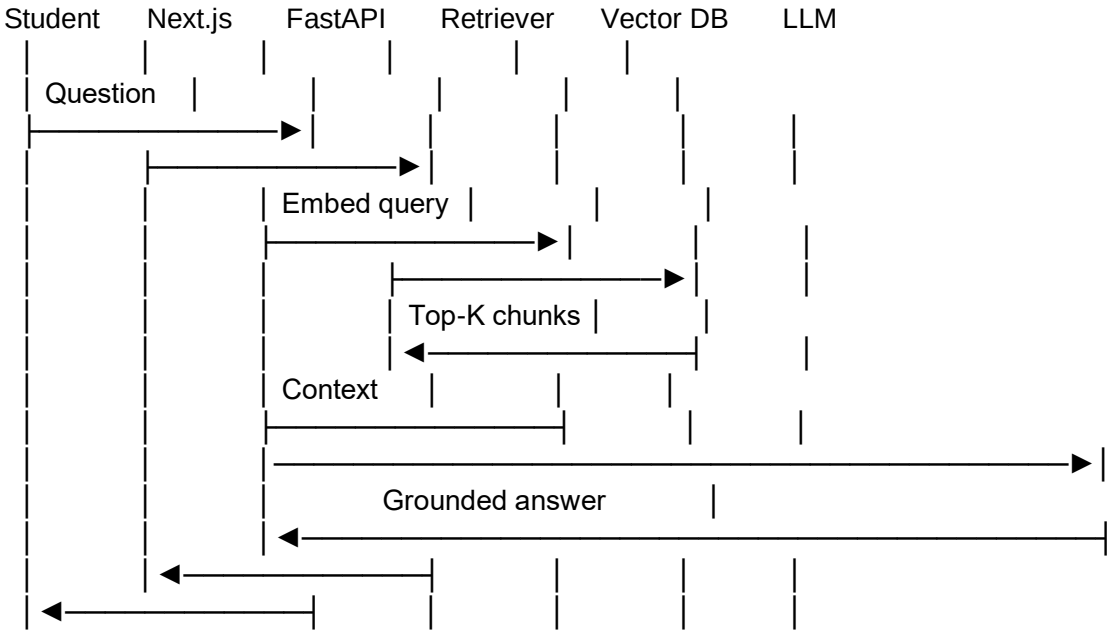


3. Component Diagram



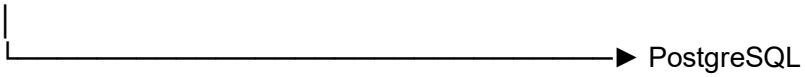


4. RAG Sequence Diagram — Question Answering



5. Document Ingestion Sequence

Student → Next.js → FastAPI → Markdown Parser → Chunker → Embedding Model → Vector DB

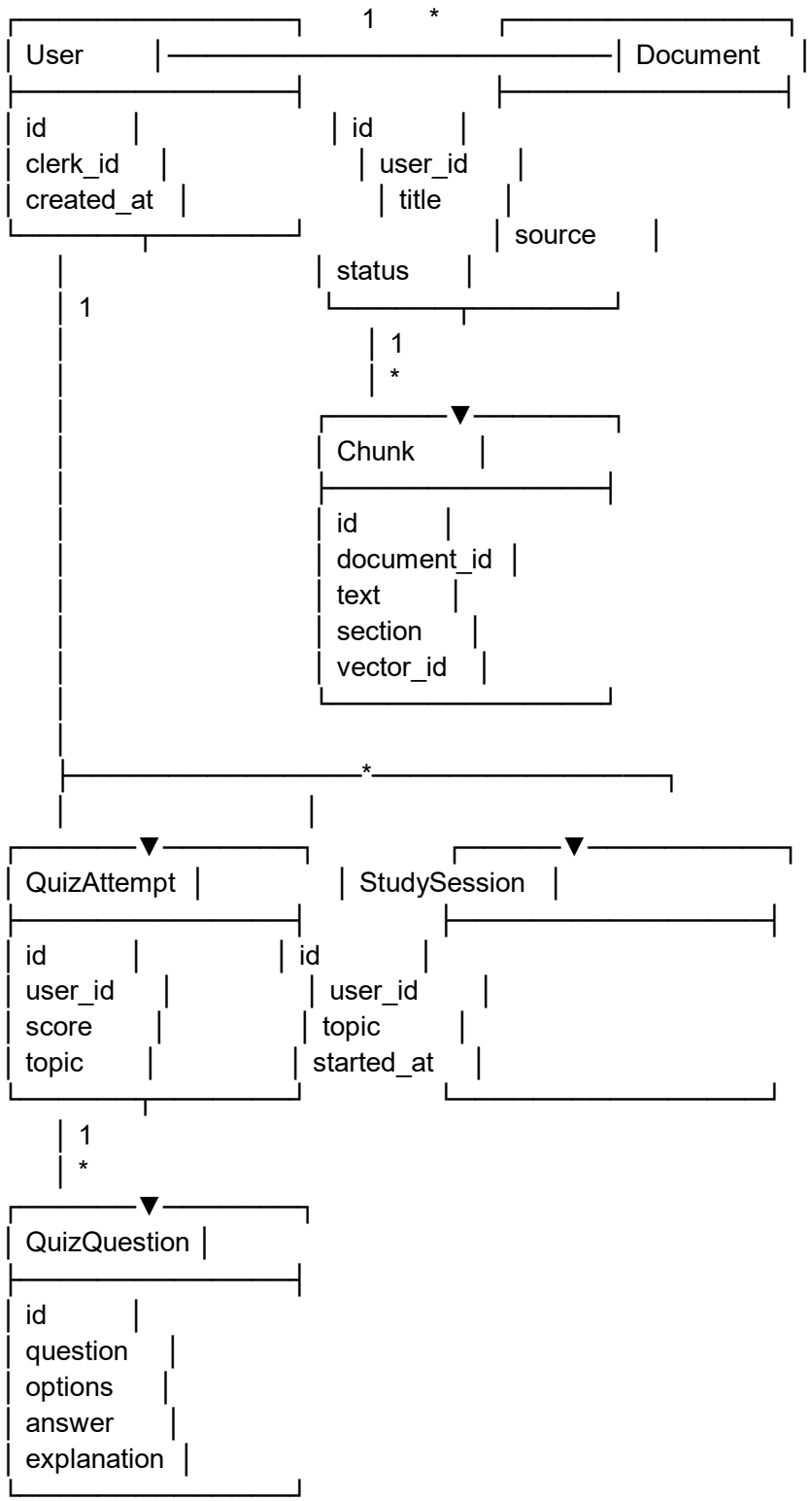


(document metadata)

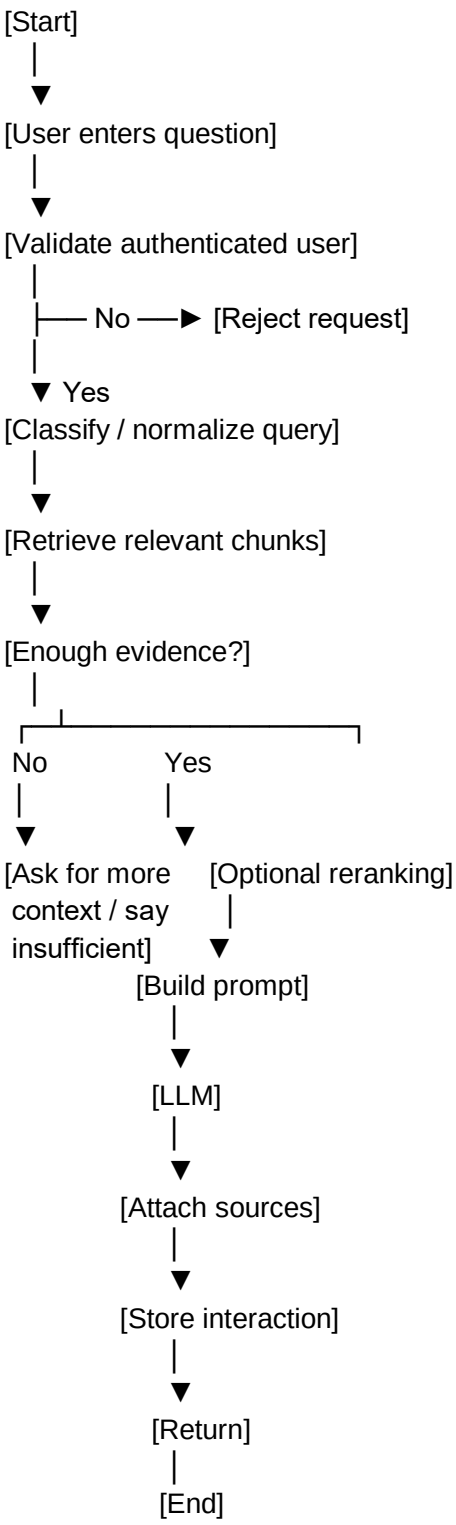
Flow:

1. User uploads/imports Markdown.
2. Backend validates ownership and file type.
3. Markdown is parsed.
4. Headings/sections are retained as metadata.
5. Content is split into retrieval-friendly chunks.
6. Embeddings are generated.
7. Chunks + embeddings + metadata are indexed.
8. Document metadata/status is stored in PostgreSQL.

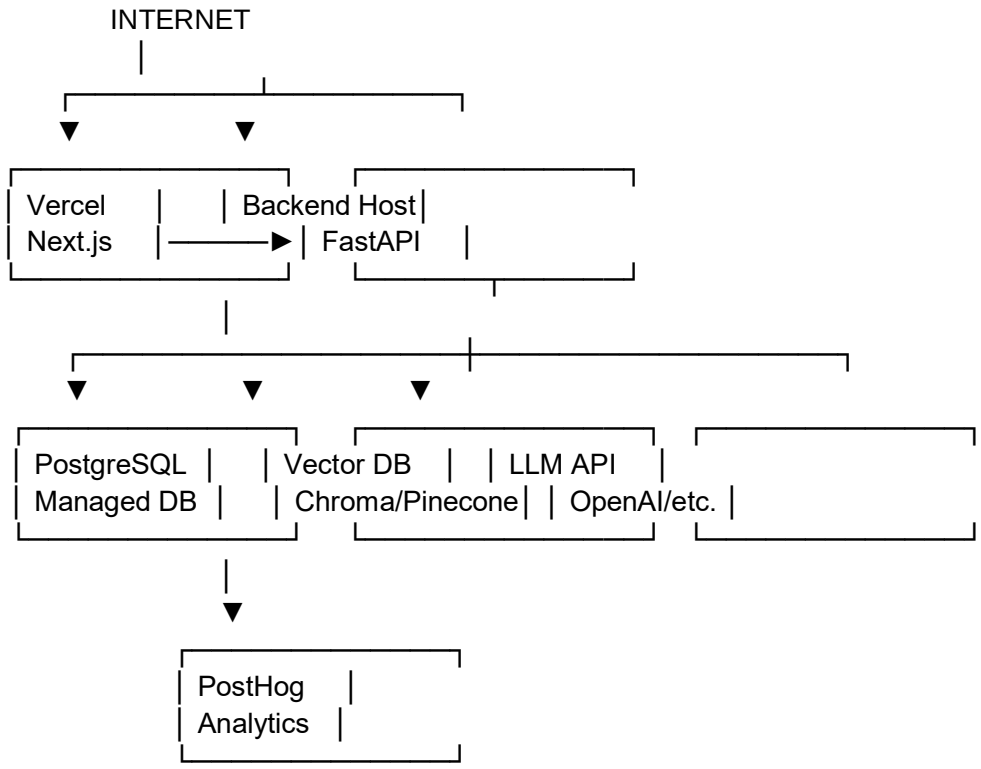
6. Class Diagram — Domain Model



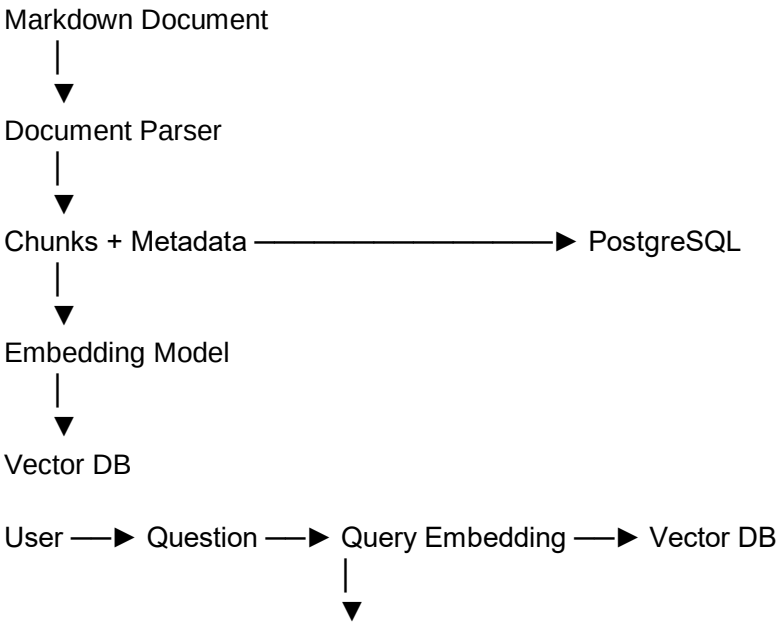
7. Activity Diagram — Ask a Question

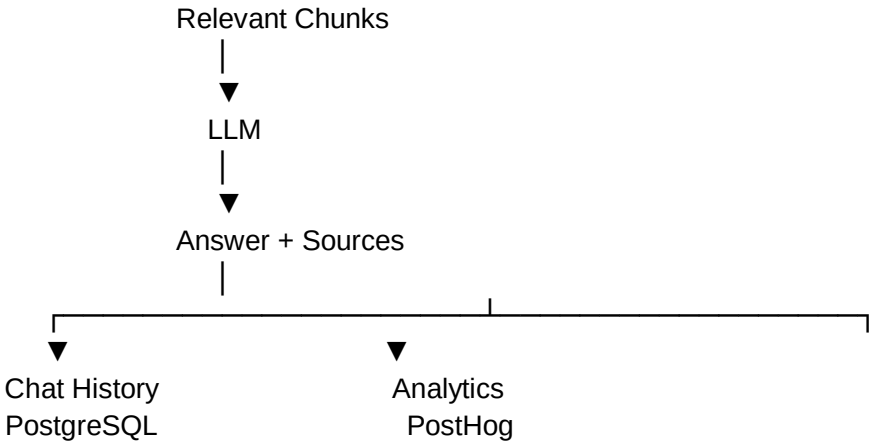


8. Deployment Diagram

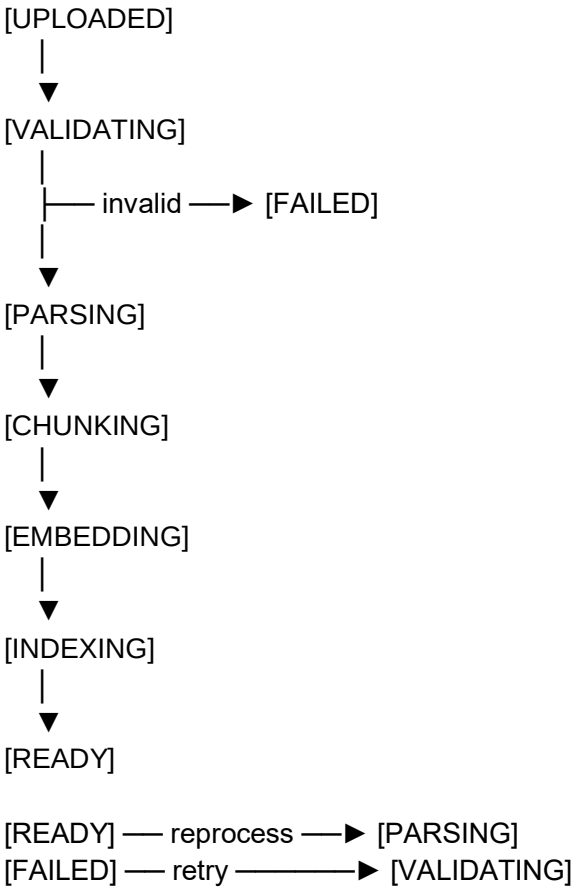


9. Database / Data Flow Diagram

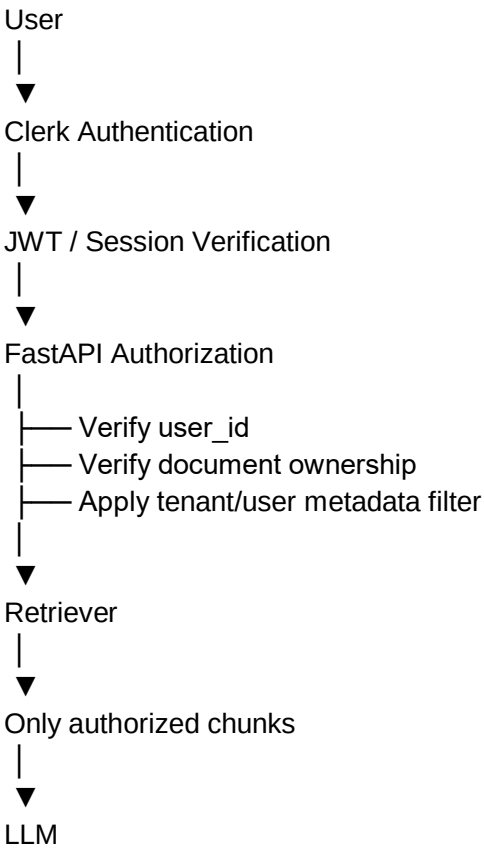




10. State Diagram — Document Processing

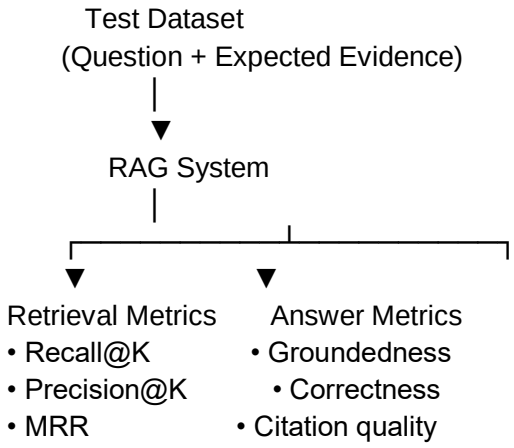


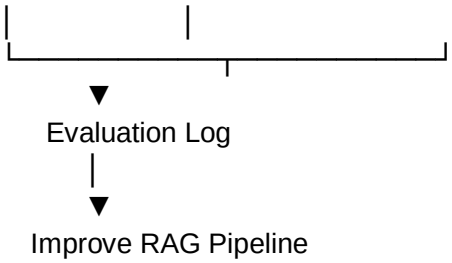
11. Security / Authorization Flow



Important: user isolation must happen before retrieval results are exposed to the LLM or returned to the client.

12. Suggested Evaluation Diagram





13. Diagram Set to Include in the Final Project Report

1. System Context / High-Level Architecture
2. Use Case Diagram
3. Component Diagram
4. RAG Question-Answer Sequence Diagram
5. Document Ingestion Sequence Diagram
6. Class / Domain Model Diagram
7. Activity Diagram
8. Deployment Diagram
9. Database / Data Flow Diagram
10. Document Processing State Diagram
11. Security / Authorization Flow
12. RAG Evaluation Flow

14. Mermaid Source Recommendation

For the actual implementation/report, redraw these diagrams in Mermaid, draw.io, or PlantUML. The ASCII diagrams above define the intended architecture and relationships; they are a blueprint, not the final visual UML notation.